DIFFERENTIAL EQUATIONS

LECTURE NOTES

WEEK XII

LAPLACE TRANSFORM

Definition Let f be a function of t, defined $t>0$. The Laplace transform of f, which is denoted by $\mathcal{L}\{f(t)\}$ or $F(s)$ is defined by

$$\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st} f(t)\,dt \left(:= \lim_{r\to\infty} \int_0^r e^{-st} f(t)\,dt \right)$$

Ex 1 $f(t)=1$, $t>0$

$$\mathcal{L}\{1\} = \int_0^\infty e^{-st}\cdot 1\,dt = \lim_{r\to\infty} \left[\int_0^r e^{-st}\,dt \right] = \lim_{r\to\infty} \frac{e^{-st}}{-s}\Big|_0^r$$

$$= \lim_{r\to\infty} \left[\frac{e^{-sr}}{-s} - \frac{e^{-s\cdot 0}}{-s} \right] = \lim_{r\to\infty} \left[\frac{e^{-sr}}{-s} + \frac{1}{s} \right] = \frac{1}{s}, \quad s>0$$

if $s>0$

$s\leq 0$ the integral diverges

$$\boxed{\mathcal{L}\{1\} = \frac{1}{s}}$$

$$\left[\ s=0 \quad \mathcal{L}\{1\} = \int_0^\infty e^{-0\cdot t}\, 1\,dt = \int_0^\infty dt = \lim_{r\to\infty} \int_0^r dt = \lim_{r\to\infty} r = \infty \text{ div.}\right.$$

Ex 2 $f(t)=e^{at}$ $t>0$

$$\mathcal{L}\{e^{at}\} = \int_0^\infty e^{-st} e^{at}\,dt = \lim_{r\to\infty} \int_0^r e^{(a-s)t}\,dt$$

$a-s<0$
$s>a$

$$= \lim_{r\to\infty} \frac{e^{(a-s)t}}{a-s}\Big|_0^r = \lim_{r\to\infty} \left[\frac{e^{(a-s)r}}{a-s} - \frac{1}{a-s} \right] = \begin{cases} \frac{1}{s-a} & \text{converges if} \\ \infty & \text{diverges otherwise} \end{cases}$$

$$\boxed{\mathcal{L}\{e^{at}\} = \frac{1}{s-a} \quad s>a}$$

Ex 2 $f(t)=t \quad t>0$

$$\mathcal{L}\{t\} = \int_0^\infty e^{-st}\, t \, dt = \lim_{r\to\infty} \int_0^r t e^{-st} dt \qquad \begin{matrix} u=t & e^{-st}dt = dv \\ du = dt & \frac{e^{-st}}{-s} = v \end{matrix}$$

$$= \lim_{r\to\infty} \left[\frac{e^{-st}}{-s} t \Big|_0^r + \frac{1}{s} \int_0^r e^{-st} dt \right] = \lim_{r\to\infty} \left[\frac{e^{-sr}}{-s} \cdot r + \frac{1}{s}\left(\frac{e^{-st}}{-s} \right)\Big|_0^r \right]$$

$$= \lim_{r\to\infty} \left[\frac{e^{-sr}}{-s} \cdot r - \frac{1}{s^2} (e^{-sr} - 1) \right] = \begin{cases} \text{converges } \frac{1}{s^2} \text{ if } s>0 \\ \text{diverges if } s \le 0 \end{cases}$$

$$\boxed{\mathcal{L}\{t\} = \frac{1}{s^2} \quad s>0}$$

Ex 3 $f(t) = \sin(at) \quad t>0$

$$\mathcal{L}\{\sin(at)\} = \int_0^\infty e^{-st} \sin(at)\, dt = \lim_{r\to\infty} \int_0^r e^{-st} \sin(at)\, dt$$

$$\begin{matrix} u = e^{-st} & \sin(at)\,dt = dv \\ du = -s e^{-st} dt & \frac{-\cos(at)}{a} = v \end{matrix}$$

$$= \lim_{r\to\infty} \left[-\frac{1}{a} e^{-st} \cos at \Big|_0^r - \frac{s}{a} \int_0^r e^{-st} \cos at \, dt \right]$$

$$\begin{matrix} u = e^{-st} & \cos at \, dt = dv \\ du = -s e^{-st} dt & \frac{\sin at}{a} = v \end{matrix}$$

$$= \lim_{r\to\infty} \left[-\frac{e^{-sr}}{a} \cos(ar) + \frac{1}{a} - \frac{s}{a} \left[e^{-st} \frac{\sin at}{a} \Big|_0^r + \frac{s}{a} \int_0^r e^{-st} \sin at \, dt \right] \right]$$

$$= \lim_{r\to\infty} \left\{ \frac{e^{-sr}}{a} \cos(ar) + \frac{1}{a} - \frac{s}{a} \left(e^{-sr} \frac{\sin ar}{a} + \frac{s}{a} \int_0^r e^{-st} \sin at\, dt \right) \right\}$$

$$= \frac{1}{a} - \frac{s^2}{a^2} \underbrace{\lim_{r\to\infty} \int_0^r e^{-st} \sin at \, dt}_{\mathcal{L}\{\sin at\}}$$

$$\mathcal{L}\{\sin at\} = \frac{1}{a} - \frac{s^2}{a^2} \mathcal{L}\{\sin at\} \qquad \frac{a^2+s^2}{a^2} \mathcal{L}\{\sin at\} = \frac{1}{a}$$

$$\boxed{\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2} \quad s>0}$$

Similarly, one can obtain

$$\boxed{\mathcal{L}\{\cos at\} = \frac{s}{s^2+a^2} \quad s>0}$$ exercise.

I. WAY

$$\mathcal{L}\{\cos at\} = \int_0^\infty e^{-st}\cos at\,dt = \lim_{r\to\infty}\int_0^r e^{-st}\cos at\,dt$$

$u = e^{-st}$ $\quad \cos at\,dt = dv$

$du = -se^{-st}dt$ $\quad \frac{\sin at}{a} = v$

$$= \lim_{r\to\infty}\left[\frac{1}{a}e^{-st}\sin at\Big|_0^r + \frac{s}{a}\int_0^r e^{-st}\sin at\,dt\right]$$

$u = e^{-st}$ $\quad \sin at\,dt = dv$

$du = -se^{-st}dt$ $\quad \frac{-\cos at}{a} = v$

$$= \lim_{r\to\infty}\left[\frac{1}{a}e^{-st}\sin at\Big|_0^r + \frac{s}{a}\left[-\frac{e^{-st}}{a}\cos at\Big|_0^r - \frac{s}{a}\int_0^r \frac{\cos at}{a}e^{-st}dt\right]\right]$$

$$= \lim_{r\to\infty}\left\{\left[\frac{1}{a}\underbrace{e^{-sr}\sin ar}_{\to 0\ s>0} - 0\right] + \frac{s}{a}\left[\underbrace{-\frac{e^{-sr}}{a}\cos ar}_{0} + \frac{1}{a}\right]\right\}$$

$$- \frac{s^2}{a^2}\underbrace{\lim_{r\to\infty}\int_0^r \frac{\cos(at)}{a}e^{-st}dt}_{\mathcal{L}\{\cos at\}}$$

$$\mathcal{L}\{\cos at\} + \frac{s^2}{a^2}\mathcal{L}\{\cos at\} = \frac{s}{a^2}$$

$$\frac{a^2+s^2}{a^2}\mathcal{L}\{\cos at\} = \frac{s}{a^2}$$

$$\mathcal{L}\{\cos at\} = \frac{s}{a^2+s^2}.$$

II. WAY For integration by parts $\quad e^{-st}dt = dv \quad v = \frac{e^{-st}}{-s}$

$\cos at = u \quad du = -a\sin at$

also can be used.

Remark Suppose f_1 and f_2 are two functions whose Laplace transforms exist for $s>a_1$ and $s>a_2$, respectively. Then, for $s>\max\{a_1,a_2\}$

$$\mathcal{L}\{c_1f_1+c_2f_2\} = c_1\,\mathcal{L}\{f_1(t)\} + c_2\,\mathcal{L}\{f_2(t)\}. \qquad s>\max\{a_1,a_2\}$$

Thus $\mathcal{L}$ is a linear operator.

Proof $\mathcal{L}\{c_1f_1+c_2f_2\} = \int_0^\infty e^{-st}(c_1f_1+c_2f_2)\,dt$

$$= c_1\underbrace{\int_0^t e^{-st}f_1\,dt}_{\mathcal{L}\{f_1\}} + c_2\underbrace{\int_0^\infty e^{-st}f_2\,dt}_{\mathcal{L}\{f_2\}}$$

$\sin 2x = 2\sin x\cos x$

$\cos 2x = 1-2\sin^2 x$

Ex (5) $\mathcal{L}\{\sin^2 at\} = \mathcal{L}\left\{\frac{1}{2} - \frac{1}{2}\cos 2at\right\}$ $\quad s>0=a_2$

$$= \frac{1}{2}\mathcal{L}\{1\} - \frac{1}{2}\mathcal{L}\{\cos 2at\}$$

$\longrightarrow s>0=a_1$

$$\mathcal{L}\{\sin^2 at\} = \frac{1}{2}\frac{1}{s} - \frac{1}{2}\frac{s}{s^2+4a^2} \qquad s>0$$

$\longrightarrow s>\max\{0,0\}$

Thm (1) Suppose that

(1) f is piecewise continuous in an interval $0\le t\le A$ for every positive A

(2) $|f(t)|\le Ke^{at}$ when $t>M$ where K, a and M are real constants with K and M are positive

Then, the Laplace Transform $\mathcal{L}\{f(t)\} = F(s) = \int_0^\infty e^{-st}f(t)\,dt$ for $s>a$. Moreover, the functions satisfying these conditions are called piecewise continuous and of exponential order e^{at} as $t\to\infty$

Thm ② (Translation Property)

Let f be such that $\mathcal{L}\{f(t)\}=F(s)$ exists when $s>s_0$. Then, for any constant a

$$\boxed{\mathcal{L}\{e^{at}f(t)\}=F(s-a)} \qquad \mathcal{L}\{f(t)\}=F(s)$$

Proof $F(s)=\int_0^\infty e^{-st}f(t)\,dt \qquad F(s-a)=\int_0^\infty e^{-(s-a)t}f(t)\,dt.$

$$=\int_0^\infty e^{-st}\cdot\underbrace{e^{at}f(t)}_{g(t)}\,dt=\mathcal{L}\{g(t)\}=\mathcal{L}\{e^{at}f(t)\}$$

Ex ① $\mathcal{L}\{e^{at}\cos bt\}=F(s-a)$ where $F(s)=\mathcal{L}\{\cos bt\}=\frac{s}{s^2+b^2}$ $s>0$

$$\mathcal{L}\{e^{at}\cos bt\}=F(s-a)=\frac{s-a}{(s-a)^2+b^2} \qquad s-a>0$$

② $\mathcal{L}\{e^{3t}\cdot t\}=F(s-3)$ where $F(s)=\mathcal{L}\{t\}=\frac{1}{s^2}$ $s>0$

$$=\frac{1}{(s-3)^2} \quad s>3. \qquad F(s-3)=\frac{1}{(s-3)^2} \quad s-3>0$$

Thm ③

Suppose f is a function satisfying the condition of Thm ① with Laplace transform $F(s)$. Then,

$$\boxed{\mathcal{L}\{t^n f(t)\}=(-1)^n\frac{d^n}{ds^n}F(s)} \qquad n=0,1,2,\dots \quad \mathcal{L}\{f(t)\}=F(s)$$

Proof $\mathcal{L}\{f(t)\}=F(s)=\int_0^\infty e^{-st}f(t)\,dt$ Diff both sides w.r.t. s, n times.

$$F'(s)=\int_0^\infty -t\,e^{-st}f(t)\,dt=(-1)^1\mathcal{L}\{tf(t)\}$$

$$\Rightarrow \mathcal{L}\{tf(t)\}=(-1)^1\underbrace{F'(s)}_{\frac{dF}{ds}}$$

$$F''(s)=\int_0^\infty (-1)^2t^2e^{-st}f(t)\,dt=(-1)^2\mathcal{L}\{t^2f(t)\} \Rightarrow \mathcal{L}\{t^2f(t)\}=(-1)^2F''(s)$$

$\vdots$

$$F^{(n)}(s)=(-1)^n\int_0^\infty e^{-st}t^n e^{-st}f(t)\,dt=(-1)^n\mathcal{L}\{t^nf(t)\} \Rightarrow \mathcal{L}\{t^nf(t)\}=(-1)^nF^{(n)}(s)$$

Ex $\mathcal{L}\{t^2 \sin at\} = (-1)^2 \frac{d^2}{ds^2} F(s)$ where $F(s) = \mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$.

$F'(s) = \frac{2as}{(s^2+a^2)^2}$ $\quad F''(s) = \frac{2a(s^2+a^2)^2 - 2(s^2+a^2)\,2s\,2as}{(s^2+a^2)^4}$

Ex $\mathcal{L}\{e^{at}t^2\} = F(s-a)$

I. WAY Use thm ① $\mathcal{L}\{e^{at}t^2\} = F(s-a)$ where

$\mathcal{L}\{t^2\} = F(s)$ using table $\mathcal{L}\{t^2\} = \frac{2}{s^3}$ so.

$\mathcal{L}\{e^{at}t^2\} = \frac{2}{(s-a)^3} = F(s-a)$

II. WAY Use thm ② $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$ $\quad F(s) = \mathcal{L}\{f(t)\}$

$f(t) = e^{at}$ $\quad \mathcal{L}\{e^{at}\} = \frac{1}{s-a} = F(s)$

$\mathcal{L}\{t^2 f(t)\} = (-1)^2 F''(s)$

$F'(s) = -\frac{1}{(s-a)^2}$ $\quad F'' = \frac{2}{(s-a)^3}$ $\quad \mathcal{L}\{t^n f(t)\} = \frac{2}{(s-a)^3}$.

Since $\mathcal{L}\{t^n f(t)\} = F(s-a)$ for $F(s) = \mathcal{L}\{t^2\}$ then

we can obtain $F(s-a) = \frac{2}{(s-a)^3}$ so $F(s) = \frac{2}{s^3}$ $s > a$

Ex $\mathcal{L}\{t^n\} = ?$

$\mathcal{L}\{t^2 \cdot 1\} = (-1)^2 \cdot \frac{d^2 F}{ds^2}$ $\quad F(s) = \mathcal{L}\{1\} = \frac{1}{s}$ $s > 0$.

$F'(s) = -\frac{1}{s^2}$ $\quad F''(s) = \frac{2}{s^3} = (-1)^2 \frac{2!}{s^{2+1}}$

$\mathcal{L}\{t^2\} = \frac{2}{s^3}$

$F'''(s) = (-1)^3 \cdot \frac{3!}{s^{3+1}}$

$\mathcal{L}\{t^3\} = (-1)^3 \cdot \frac{d^3 F}{ds^3} = -\frac{3!}{s^4}$ $\quad F^{IV}(s) = (-1)^4 \frac{3!\,4}{s^5}$

$\mathcal{L}\{t^4\} = (-1)^4 \frac{d^4 F}{ds^4} = \frac{4!}{s^5}$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \quad s > 0$$

Thm. 4 (TRANSFORMATIONS OF DERIVATIVES)

Let $f(t)$ be a function satisfying the conditions of Thm 1. Then, $\mathcal{L}\{f'(t)\}$ exists for all $s>a$ and

$$\boxed{\mathcal{L}\{f'(t)\} = s\,\mathcal{L}\{f(t)\} - f(0)}$$

Remark $\mathcal{L}\{f''(t)\} = \mathcal{L}\{(\underbrace{f'(t)}_{g(t)})'\} = \mathcal{L}\{g'(t)\} = s\cdot\mathcal{L}\{g(t)\} - g(0)$

$$= s\cdot\mathcal{L}\{f'(t)\} - f'(0) = s\,[s\,\mathcal{L}\{f(t)\} - f(0)] - f'(0)$$

$$\mathcal{L}\{f''(t)\} = s^2\mathcal{L}\{f(t)\} - s f(0) - f'(0)$$

$\vdots$

$$\mathcal{L}\{f^{(n)}(t)\} = s\,\mathcal{L}\{f^{(n-1)}(t)\} - f^{(n-1)}(0)$$

$$\boxed{\mathcal{L}\{f^{(n)}(t)\} = s^n\mathcal{L}\{f(t)\} - s^{n-1}f(0) - s^{n-2}f'(0) - s^{n-3}f''(0) - \cdots - s f^{(n-2)}(0) - f^{(n-1)}(0)}$$

Solutions of IVP's

Ex. 1 Solve $y''(t) - y'(t) - 6y(t) = 2 \qquad y(0)=1 \quad y'(0)=0$

Usual way $r^2 - r - 6 = 0 \quad (r-3)(r+2)=0 \quad r=3 \quad r=-2 \quad y_h = c_1 e^{3t} + c_2 e^{-2t}$

y_p $\quad y(t) = A t^0 \quad \underbrace{A''}_{0} - \underbrace{A'}_{0} - 6A = 2 \qquad A = -1/3$

$$y(t) = y_h(t) + y_p(t) = c_1 e^{3t} + c_2 e^{-2t} - 1/3$$

$y(0) = 1 = c_1 + c_2 - 1/3 \qquad y'(0) = 0 = 3c_1 - 2c_2 = 0 \Rightarrow c_1 = 2/3\, c_2$

$$\boxed{y(t) = \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t} - \frac{1}{3}}$$

Apply Laplace $\mathcal{L}\{y''\} - \mathcal{L}\{y'\} - 6\mathcal{L}\{y\} = \mathcal{L}\{1\} \qquad F(s) = \mathcal{L}\{y\}$

$\mathcal{L}\{y''\} = s^2\mathcal{L}\{y\} - s y(0) - y'(0) \qquad \mathcal{L}\{1\} = \frac{1}{s}$

$\mathcal{L}\{y'\} = s\,\mathcal{L}\{y\} - y(0)$

$$\mathcal{L}\{y''\} - \mathcal{L}\{y'\} - 6\mathcal{L}\{y\} = s^2\mathcal{L}\{y\} - s y(0) - y'(0) - s\mathcal{L}\{y\} + y(0) - 6\mathcal{L}\{y\} = s^2\mathcal{L}\{y\} - s - s\mathcal{L}\{y\} + 1 - 6\mathcal{L}\{y\}$$

$y(0) = 1$

$y'(0) = 0$

$$s^2 \mathcal{L}\{y\} - s - s\,\mathcal{L}\{y\} + 1 - 6\,\mathcal{L}\{y\} = \frac{2}{s}$$

$$(s^2 - s - 6)\,\mathcal{L}\{y\} = \frac{2}{s} + s - 1$$

$$\mathcal{L}\{y\} = \frac{s^2 - s + 2}{s(s-3)(s+2)} = \frac{A}{s} + \frac{B}{s-3} + \frac{C}{s+2}$$

$(s-3)(s+2)$ $(s+2).s$ $(s-3).s$

$A = -1/3$ $B = 8/15$ $C = 4/5$

$$\mathcal{L}\{y\} = -\frac{1}{3}\cdot\underbrace{\frac{1}{s}}_{1} + \frac{8}{15}\cdot\underbrace{\frac{1}{s-3}}_{e^{3t}} + \frac{4}{5}\cdot\underbrace{\frac{1}{s+2}}_{e^{-2t}}$$

$$y(t) = -\frac{1}{3}\cdot 1 + \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t}$$

Ex 7 $y' - y = e^{3t}$ $y(0) = 2$ solve using Laplace transform

$$\mathcal{L}\{y' - y\} = \mathcal{L}\{e^{3t}\}$$

$$\mathcal{L}\{y'\} - \mathcal{L}\{y\} = \frac{1}{s-3}$$ → leave alone $Y(s) = \mathcal{L}\{y(t)\}$

$$s\,\mathcal{L}\{y\} - \underbrace{y(0)}_{2} - \mathcal{L}\{y\} = \frac{1}{s-3}$$

$$(s-1)\,\mathcal{L}\{y\} = \frac{1}{s-3} + 2 \qquad \mathcal{L}\{y\} = \frac{2s-5}{(s-3)(s-1)} = \frac{A}{(s-1)} + \frac{B}{(s-3)}$$

$(s-3)$ $(s-1)$

$$A + B = 2$$
$$-3A - B = -5 \Rightarrow A = \frac{3}{2} \quad B = \frac{1}{2}$$

$$\mathcal{L}\{y\} = \frac{3}{2}\cdot\frac{1}{s-1} + \frac{1}{2}\,\frac{1}{s-3} \Rightarrow \frac{3}{2}e^{t} + \frac{1}{2}e^{3t} = y(t)$$

$$\mathcal{L}\{y\} = F(s) \qquad y(t) = \mathcal{L}^{-1}\{F(s)\}$$

Remark ① To find solution of I.V.P, we first find $F(s) = \mathcal{L}\{y\}$ then we apply inverse Laplace transform $\mathcal{L}^{-1}\{F(s)\}$ to find $y(t)$.

② When we apply Laplace Transform to a D.E, we end up with an algebraic function no more derivatives

Ex 7. $y'' - y' - 2y = 18\, e^{-t} \sin 3t \qquad y(0) = 0 \qquad y'(0) = 3$

Solve by Laplace. Let $Y(s) = \mathcal{L}\{y(t)\}$

$$s^2 Y(s) - s\underbrace{y(0)}_{0} - \underbrace{y'(0)}_{3} - sY(s) + \underbrace{y(0)}_{0} - 2Y(s) = 18\, \mathcal{L}\{e^{-t}\sin 3t\}$$

$$\mathcal{L}\{e^{-t}\sin 3t\} = F(s-(-1)) = F(s+1) \text{ where } F(s) = \mathcal{L}\{\sin 3t\} = \frac{3}{s^2+9}$$

$$F(s+1) = \frac{3}{(s+1)^2+9}$$

$$(s^2 - s - 2)\, Y(s) - 3 = 18 \cdot \frac{3}{(s+1)^2+9}$$

$$Y(s) = \frac{\frac{54}{(s+1)^2+9} + 3}{(s^2-s-2)} = \frac{54}{((s+1)^2+9)(s^2-s-2)} + \frac{3}{s^2-s-2}$$

take inv Laplac

$$Y(s) = \frac{54 + 3((s+1)^2+9)}{((s+1)^2+9)\,(s-2)(s+1)}$$

$$Y(s) = \frac{3s^2 + 6s + 84}{((s+1)^2+9)\,(s-2)(s+1)} = \frac{A}{(s+1)} + \frac{B}{s-2} + \frac{Cs+D}{(s+1)^2+9}$$

$(s-2)((s+1)^2+9) \qquad (s+1)((s+1)^2+9) \qquad (s-2)(s+1)$

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}^{-1}\left\{\frac{-3}{s+1} + \frac{2}{s-2} + \frac{s-3}{(s+1)^2+9}\right\} = y(t)$$

$$y(t) = -3e^{-t} + 2e^{2t} + \mathcal{L}^{-1}\left\{\frac{s-3}{(s+1)^2+9}\right\}$$

$$\frac{s-3}{(s+1)^2+9} = \frac{s+1-1-3}{(s+1)^2+9} = \frac{s+1}{(s+1)^2+9} - \frac{4}{3}\,\frac{1 \cdot 3}{(s+1)^2+9}$$

$\mathcal{L}\{\sin at\} = \frac{a}{s^2+a^2}$

$e^{-t}\sin 3t.$

$$\mathcal{L}^{-1}\left\{\frac{s-3}{(s+1)^2+9}\right\} = e^{-t}\cos 3t - \frac{4}{3} e^{-t}\sin 3t$$

$$y(t) = -3e^{-t} + 2e^{2t} + e^{-t}\cos 3t - \frac{4}{3}e^{-t}\sin 3t$$

Ex 7 $y''' - 5y'' + 7y' - 3y = 20\sin t$ $\quad y(0)=0 \quad y'(0)=0 \quad y''(0)=-2$

Use Laplace transform for solution. Let $Y(s) = \mathcal{L}\{y(t)\}$

$$\underbrace{s^3Y(s) - s^2y(0) - sy'(0) - y''(0)}_{y'''} - 5[\underbrace{s^2Y(s) - sy(0) - y'}_{y''}]$$

$$+7[sY(s) - y(0)] - 3Y(s) = 20 \cdot \frac{1}{s^2+1}$$

$$(s^3 - 5s^2 + 7s - 3)Y(s) = \frac{20}{s^2+1} - 2$$

$$\begin{array}{r|l} s^3 - 5s^2 + 7s - 3 & s-1 \\ \hline s^3 - s^2 & s^2 - 4s + 3 \\ -4s^2 + 7s & \\ -4s^2 + 4s & \\ 3s - 3 & \end{array}$$

$$Y(s) = \frac{-2s^2 + 18}{(s^2+1)(s^3 - 5s^2 + 7s - 3)}$$

$$= \frac{-2s^2 + 18}{(s^2+1)(s-1)^2(s-3)}$$

$$Y(s) = \frac{-2(\cancel{s-3})(s+3)}{(s^2+1)(s-1)^2(\cancel{s-3})} \qquad Y(s) = \frac{-2(s+3)}{(s^2+1)(s-1)^2}$$

$$Y(s) = \frac{\overset{-3}{A}s + \overset{1}{B}}{s^2+1} + \frac{C = -4}{(s-1)^2} + \frac{D = 3}{s-1}$$

$$\mathcal{L}^{-1}\{Y(s)\} = y(t) = \mathcal{L}\left\{\frac{-3s+1}{s^2+1}\right\} - 4\mathcal{L}\left\{\frac{1}{(s-1)^2}\right\} + 3\mathcal{L}\left\{\frac{1}{s-1}\right\}$$

$$\frac{-3s+1}{s^2+1} = -3\underbrace{\frac{s}{s^2+1}}_{\cos t} + \underbrace{\frac{1}{s^2+1}}_{\sin t}$$

$$y(t) = -3\cos t + \sin t - 4te^t + 3e^t$$

GAMMA FUNCTION

- $\Gamma(p)$ $\quad p \in \mathbb{R} \quad p > 0$
- $\Gamma(p) = \int_0^\infty e^{-t}\, t^{p-1}\, dt$, $p > 0$, $p \in \mathbb{R}$ is a special function called Gamma Function
- $\Gamma(p+1) = \int_0^\infty e^{-t}\, t^p\, dt = \mathcal{L}\{t^p\}$ when $s = 1$

Note that $\mathcal{L}\{t^p\} = \int_0^\infty e^{-st}\, t^p\, dt$

Properties

(1) $\boxed{\Gamma(p+1) = p\,\Gamma(p)}$

(2) $\Gamma(1) = \int_0^\infty e^{-t}\, dt = \lim_{n\to\infty} \int_0^n e^{-t}\, dt = \lim_{n\to\infty} \left.\frac{e^{-t}}{-1}\right|_0^n = \lim_{n\to\infty} (-\cancel{e^{-n}}^{\,0} + 1) = 1$

$\Gamma(1) = 1$

(3) $p = n \in \mathbb{N}$ $\quad \Gamma(n+1) = n\,\Gamma(n) = n(n-1)\,\Gamma(n-1) = n(n-1)\ldots \quad 2\cdot 1\cdot \underbrace{\Gamma(1)}_{1}$

$\boxed{\Gamma(n+1) = n!}$

(4) $\boxed{\mathcal{L}\{t^p\} = \frac{\Gamma(p+1)}{s^{p+1}} \quad \text{for } p > -1 \quad p \in \mathbb{R}}$

For example $\quad \mathcal{L}\{t^{1/3}\} = \frac{\Gamma(1/3+1)}{s^{1/3+1}} = \frac{\Gamma(4/3)}{s^{4/3}}$

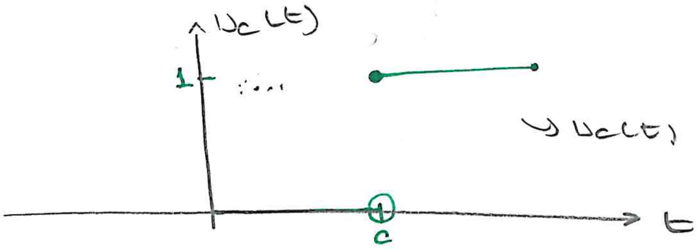
$\mathcal{L}\{t^2\} = \frac{\Gamma(2+1)}{s^{2+1}} = \frac{\Gamma(3)}{s^3} = \frac{2!}{s^3} = \frac{2}{s^3}$

$n \in \mathbb{Z}$ $\quad \boxed{\mathcal{L}\{t^n\} = \frac{\Gamma(n+1)}{s^{n+1}} = \frac{n!}{s^{n+1}}}$

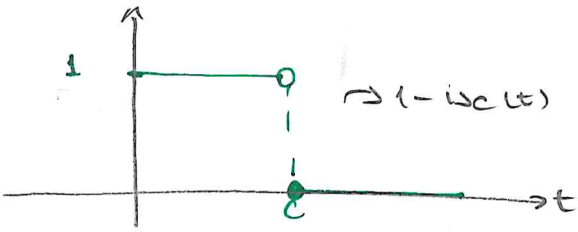
STEP FUNCTION

Definition The function (Heaviside function) $u_c(t) = \begin{cases} 0, & t < c \\ 1, & t \geq c \end{cases}$ $c > 0$ is called

unit step function



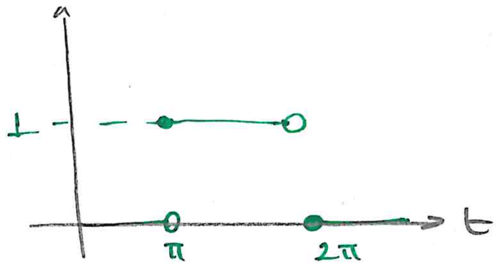
$$u_c(t) = \begin{cases} 0 & t < c \\ 1 & t \geq c \end{cases}$$



$$1 - u_c(t) = \begin{cases} 1 & t < c \\ 0 & t \geq c \end{cases}$$

Ex 7 Sketch $h(t) = u_\pi(t) - u_{2\pi}(t)$

$$h(t) = \begin{cases} 0 - 0 = 0 & t < \pi \\ 1 - 0 = 1 & \pi \leq t < 2\pi \\ 1 - 1 = 0 & t \geq 2\pi \end{cases}$$



Remark $\mathcal{L}\{u_c(t)\} = \int_0^\infty e^{-st} u_c(t)\,dt = \int_0^c e^{-st} \underbrace{u_c(t)}_{0}\,dt + \int_c^\infty e^{-st} \underbrace{u_c(t)}_{1}\,dt$

$$= \int_c^\infty e^{-st}\,dt = \lim_{r \to \infty} \int_c^r e^{-st}\,dt$$

$$= \lim_{r \to \infty} \left(\frac{e^{-st}}{-s} \Big|_c^r \right) = \lim_{r \to \infty} \left[\frac{e^{-sr}}{-s} - \frac{e^{-sc}}{-s} \right]$$

$\frac{e^{-sr}}{-s} \to 0$ if $s > 0$

$$\boxed{\mathcal{L}\{u_c(t)\} = \frac{e^{-sc}}{s}, \quad s > 0}$$

$$\mathcal{L}\{h(t)\} = \mathcal{L}\{u_\pi(t) - u_{2\pi}(t)\} = \frac{e^{-s\pi}}{s} - \frac{e^{-2s\pi}}{s}$$

Theorem Suppose $F(s)=\mathcal{L}\{f(t)\}$ exists for $s>a\geq 0$. If c is a positive constant and

$$g(t)=\begin{cases}0 & 0<t<c\\ f(t-c) & t\geq c\end{cases} = f(t-c)\cdot\underbrace{\begin{cases}0 & 0<t<c\\ 1 & t\geq c\end{cases}}_{u_c(t)}$$

$$g(t)=u_c(t)\,f(t-c)$$

Then, $\mathcal{L}\{g(t)\}=\mathcal{L}\{u_c(t)\,f(t-c)\}=e^{-cs}F(s)\quad s>a$

Ex 7 $h(t)=\begin{cases}\sin t & 0\leq t<2\pi\\ \sin t+\cos t & t\geq 2\pi\end{cases}\qquad \mathcal{L}\{h(t)\}=?$

$$=\sin t+\begin{cases}0 & 0\leq t<2\pi\\ \cos t & t\geq 2\pi\end{cases}$$

$$=\sin t+\cos t\underbrace{\begin{cases}0 & 0\leq t<2\pi\\ 1 & t\geq 2\pi\end{cases}}_{u_{2\pi}(t)}$$

$$h(t)=\sin t+\underbrace{\cos t}_{\cos(t-2\pi)}\cdot u_{2\pi}(t)$$

$\cos(t-2\pi)$ cos. is 2π periodic.

$$h(t)=\sin t+u_{2\pi}(t)\cos(t-2\pi)$$

$$\mathcal{L}\{h(t)\}=\frac{1}{s^2+1}+e^{-2\pi s}F(s)\quad \text{where}\quad F(s)=\mathcal{L}\{\cos t\}=\frac{s}{s^2+1}$$

$$\mathcal{L}\{h(t)\}=\frac{1}{s^2+1}+e^{-2\pi s}\frac{s}{s^2+1}.$$

Remark $\mathcal{L}\{u_c(t)\,f(t-c)\}=e^{-cs}F(s)$

$$\mathcal{L}^{-1}\{e^{-cs}F(s)\}=u_c(t).f(t-c)$$

Remark $f(t)=\begin{cases}g(t) & 0\leq t<a\\ h(t) & a\leq t<b\\ l(t) & t\geq b\end{cases}$ $\quad f(t)=g(t)+(h(t)-g(t))u_a(t)+(l(t)-h(t))u_b(t)$

$f(t)=\begin{cases}g(t) & 0\leq t<a\\ h(t) & t\geq a\end{cases}$ $\quad f(t)=g(t)+(h(t)-g(t))u_a(t)$

Ex Find $F(s)$ if $f(t) = \begin{cases} t & 0 \le t < 5 \\ \sin t & t \geq 5. \end{cases}$

$$f(t) = t + \begin{cases} 0 & 0 \le t < 5 \\ \sin t - t & 5 \geq t. \end{cases}$$

$$f(t) = t + (\sin t - t) \underbrace{\begin{cases} 0 & 0 \le t < 5 \\ 1 & t \geq 5 \end{cases}}_{u_5(t)}$$

$\sin(a+b) = \sin a . \cos b + \sin b . \cos a$

$$f(t) = t + u_5(t).(\sin t - t)$$

$$\sin t - t = \sin((t-5)+5) - \overset{t}{} = \sin(t-5).\cos 5 + \sin 5 \cos(t-5)$$

$$-(t-5)-5$$

$$f(t) = t + u_5(t) \sin(t-5).\cos 5 + u_5(t) \cos(t-5) \sin 5 - u_5(t)(t-5)$$
$$- 5 u_5(t)$$

$$F(s) = \frac{1}{s^2} + \cos 5 \, e^{-5s} . \frac{1}{s^2+1} + \sin 5 \, e^{-5s} \frac{s}{s^2+1} - e^{-5s} . \frac{1}{s^2} - 5 . \frac{e^{-5s}}{s}$$

Ex Find $f(t)$ if $\mathcal{L}\{f(t)\} = F(s) = \dfrac{2s \, e^{-55s}}{(s^2-2s+3)} = e^{-55s} . G(s)$

where $G(s) = \dfrac{2s}{s^2-2s+3} = \dfrac{2s}{(s-1)^2+2} = 2 \dfrac{s-1+1}{(s-1)^2+2} = 2 . \dfrac{(s-1)}{(s-1)^2+2} + 2 . \dfrac{1}{(s-1)^2+}$

$$\mathcal{L}^{-1}\{G(s)\} = 2 . \mathcal{L}^{-1}\left\{\frac{s-1}{(s-1)^2+2}\right\} + \frac{2}{\sqrt{2}} \mathcal{L}^{-1}\left\{\underbrace{\frac{\sqrt{2}}{(s-1)^2+2}}_{e^t . \sin\sqrt{2}t}\right\}$$

$$g(t) = 2 e^t . \cos(\sqrt{2}t) + \sqrt{2} e^t \sin(\sqrt{2}t)$$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\{e^{-55s} G(s)\} = u_{55}(t) g(t-55)$$

$$= u_{55}(t)\left[2 . e^{t-55} \cos(\sqrt{2}(t-55)) + \sqrt{2} e^{t-55} \sin\sqrt{2}(t-5\right.$$

Ex. $y''+y=p(t)$ where $p(t)=\begin{cases} t/2 & 0\le t<6 \\ 3 & t\geq 6 \end{cases}$

$y(0)=0 \quad y'(0)=1$

REVIEW AND SUMMARY

1. $\mathcal{L}\{f(t)\}=F(s)=\int_0^\infty e^{-st}f(t)\,dt$
2. $\mathcal{L}\{1\}=1/s \quad s>0$
3. $\mathcal{L}\{t\}=\frac{1}{s^2} \quad s>0$
4. $\mathcal{L}\{t^n\}=\frac{n!}{s^{n+1}} \quad s>0,\ n\in\mathbb{N}$
5. $\mathcal{L}\{t^p\}=\frac{\Gamma(p+1)}{s^{p+1}},\ s>0,\ p\in\mathbb{R}\ p>-1$
6. $\mathcal{L}\{e^{at}\}=\frac{1}{s-a},\ s>a$
7. $\mathcal{L}\{\sin at\}=\frac{a}{s^2+a^2},\ s>a$
8. $\mathcal{L}\{\cos at\}=\frac{s}{s^2+a^2},\ s>a$
9. $\mathcal{L}\{e^{at}f(t)\}=F(s-a)$ where $F(s)=\mathcal{L}\{f(t)\}$
10. $\mathcal{L}\{t^n f(t)\}=(-1)^n\frac{d^nF(s)}{ds^n}$ where $F(s)=\mathcal{L}\{f(t)\}$
11. $\mathcal{L}\{u_c(t)\}=\frac{e^{-cs}}{s},\ s>0$
12. $\mathcal{L}\{u_c(t)f(t-c)\}=e^{-cs}F(s) \qquad (\mathcal{L}^{-1}\{e^{-cs}F(s)\}=u_c(t)f(t-c))$
13. $\mathcal{L}\{f'(t)\}=sF(s)-f(0)$
14. $\mathcal{L}\{f^{(n)}(t)\}=s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)-\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$

Ex 7 $y'' + y = g(t)$ $\qquad g(t) = \begin{cases} t/2 & 0 \le t < 6 \\ 3 & t \ge 6 \end{cases}$ $\qquad$ Solve I.V.P

$y(0) = 0 \quad y'(0) = 1$

$$s^2 Y(s) - s\underbrace{y(0)}_{0} - \underbrace{y'(0)}_{1} + Y(s) = \mathcal{L}\{g(t)\}$$

$$(s^2+1)Y(s) = \mathcal{L}\{g(t)\} + 1$$

$$g(t) = \frac{t}{2} + \left(3 - \frac{t}{2}\right)\begin{cases} 0 & 0 \le t < 6 \\ 1 & t \ge 6 \end{cases} = \frac{t}{2} - \frac{1}{2}(t-6)\cdot u_6(t)$$

$$\mathcal{L}\{g(t)\} = \frac{1}{2}\mathcal{L}\{t\} - \frac{1}{2}\mathcal{L}\{(t-6)u_6(t)\} = \frac{1}{2}\frac{1}{s^2} - \frac{1}{2}e^{-6s}\cdot\frac{1}{s^2}$$

$$\mathcal{L}\{u_c(t) f(t-c)\} = e^{-cs}F(s)$$
$$\mathcal{L}^{-1}\{e^{-cs}F(s)\} = u_c(t) f(t-c)$$

$$(s^2+1)\cdot Y(s) = \frac{1}{2}\frac{1-e^{-6s}}{s^2} + 1$$

$$Y(s) = \frac{1}{2}\frac{1-e^{-6s}}{s^2(s^2+1)} + \frac{1}{s^2+1}$$

Take inverse Laplace to find $y(t)$

$$y(t) = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1-e^{-6s}}{s^2(s^2+1)}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\}$$

$$\frac{1}{s^2(s^2+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{Cs+D}{s^2+1} = \frac{1}{s^2} - \frac{1}{s^2+1}$$

$s(s^2+1) \quad (s^2+1) \quad (s^2)$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\} = t - \sin t$$

$$1 = As^3 + As + Bs^2 + B + Cs^3 + Ds^2$$

$$1 = \underbrace{(A+C)}_{0}s^3 + \underbrace{(B+D)}_{0}s^2 + \underbrace{A}_{0}s + \underbrace{B}_{1}$$

$$\mathcal{L}^{-1}\left\{e^{-6s}\frac{1}{s^2(s^2+1)}\right\} = u_6(t) f(t-6)$$

$A = 0 \quad B = 1$

$C = 0 \quad D = -1$

$$= u_6(t)\left[(t-6) - \sin(t-6)\right]$$

where $f(t) = \mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\} = t - \sin t$

$$y(t) = \frac{1}{2}\left[t - \sin t - u_6(t)\left[(t-6) - \sin(t-6)\right]\right] + \sin t$$

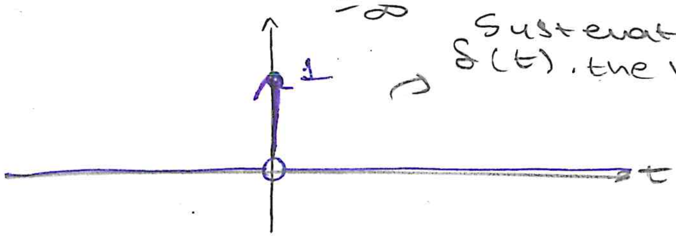
IMPULSE FUNCTION

Definition

The function $\delta(t) = \begin{cases} 0 & t \neq 0 \\ \infty & t = 0 \end{cases}$ is called impulse (Dirac De[lta]) function.

$$\int_{-\infty}^{\infty} \delta(t)\,dt = 1.$$

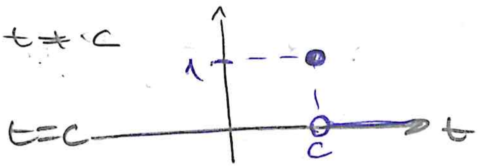
Systematic representation of $\delta(t)$



$\delta(t)$, the height of the arrow represents the area $\int_{-\infty}^{\infty} \delta(t)\,dt = 1$.

Similarly $\delta(t-c) = \begin{cases} 0 & t \neq c \\ \infty & t = c \end{cases}$

$$\int_{-\infty}^{\infty} \delta(t-c) = 1$$



PROPERTIES (1) $\mathcal{L}\{\delta(t-c)\} = e^{-cs}$ (2) $\mathcal{L}\{\delta(t-c) f(t)\} = e^{-cs} f(c)$

$\mathcal{L}\{\delta(t)\} = 1$

Proof $\mathcal{L}\{\delta(t-c) f(t)\} = \int_0^{\infty} e^{-st} \delta(t-c) f(t)\,dt$

$$= \int_{-\infty}^{\infty} e^{-st} \delta(t-c) f(t)\,dt$$

Since $\delta(t-c) = 0,\ t < c$ $(c > 0)$

$$= \int_{-\infty}^{\infty} \delta(t-c) \underbrace{e^{-st} f(t)}_{g(t)}\,dt$$

$$= g(c) = e^{-sc} f(c) \quad \#$$

Ex Find solution of $y'' + 4y = \delta(t-\pi) - \delta(t-2\pi)$ $y(0) = 0, y'(0) = 0$

$$s^2 Y(s) - s\underbrace{y(0)}_{0} - \underbrace{y'(0)}_{0} + 4Y(s) = e^{-\pi s} - e^{-2\pi s}$$

$$Y(s) = \frac{e^{-\pi s}}{s^2+4} - \frac{e^{-2\pi s}}{s^2+4}$$

$$\mathcal{L}^{-1}\{Y(s)\} = \mathcal{L}\{e^{-\pi s}\underbrace{\frac{1}{s^2+4}}_{G(s)}\} - \mathcal{L}\{e^{-2\pi s}\underbrace{\frac{1}{s^2+4}}_{G(s)}\}$$

$$\left(\mathcal{L}\{\frac{1.2}{2(s^2+4)}\} = \frac{1}{2}\underbrace{\sin 2t}_{g(t)}\right) \Rightarrow y(t) = u_\pi(t)\, g(t-\pi) - u_{2\pi}(t)\, g(t-2\pi)$$

$$y(t) = \frac{1}{2}u_\pi(t)\sin(\underbrace{2(t-\pi)}_{\sin 2t})$$

$$- \frac{1}{2}u_{2\pi}(t)\sin\underbrace{(2(t-2\pi))}_{\sin 2t}$$

Ex $y''+y = \delta(t-2\pi)\cos t$ $\quad y(0)=0 \quad y'(0)=1$ $\quad$ solve I.V.P.

$$s^2Y(s) - s\underbrace{y(0)}_{0} - \underbrace{y'(0)}_{1} + Y(s) = \underbrace{\mathcal{L}\{\delta(t-2\pi)\cos t\}}_{e^{-2\pi s}\underbrace{\cos(2\pi)}_{1}}$$

$$(s^2+1)Y(s) = e^{-2\pi s} + 1$$

take inverse Laplace.

$$Y(s) = \frac{e^{-2\pi s}}{s^2+1} + \frac{1}{s^2+1}$$

$$y(t) = u_{2\pi}(t)\underbrace{\sin(t-2\pi)}_{\sin t} + \sin t$$

$$y(t) = \sin t\,(1 + u_{2\pi}(t))$$

CONVOLUTION INTEGRAL

Definition Let f and g are two functions which are piecewise continuous on every finite interval $0 \le t \le A$ and of exponential order. The function denoted by $f * g$ (and read as f convolution g) is defined as:

$$\boxed{(f * g)(t) = \int_0^t f(u)\, g(t-u)\, du = \int_0^t f(t-u)\, g(u)\, du}$$

Properties
1. $f * g = g * f$ commutative
2. $f * (g_1 + g_2) = f * g_1 + f * g_2$ distributive
3. $(f * g) * h = f * (g * h)$ associative
4. $f * 0 = 0$.
5. $f * 1 \ne f$ (for example $\sin t * 1 = \int_0^t \sin u \cdot 1\, du$
$= -\cos u \Big|_0^t = -\cos t + 1 \ne \sin t$
6. $f * f$ may take negative values as well.

$\cos A \cdot \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$

Ex

$$\cos t * \cos t = \int_0^t \cos u\,(\cos(t-u))\,du = \frac{1}{2}\int_0^t [\cos t + \cos(2u-t)]\,du$$

$$= \frac{1}{2}\left[\cos t\; u \Big|_{u=0}^{t} + \frac{\sin(2u-t)}{2}\Big|_0^t\right].$$

$$= \frac{1}{2} t\cos t + \frac{1}{4}[\sin t - \sin(-t)]$$

$$\cos t * \cos t \Big|_{t=\pi} = -\frac{\pi}{2} < 0$$

Thm Let f and g be piecewise continuous functions on every finite interval $0 \le t \le A$ and of exponential order e^{at}.

Then,

$$\mathcal{L}\{f * g\} = F(s)G(s)$$

$$\mathcal{L}^{-1}\{F(s)G(s)\} = (f * g)(t)$$

Ex 1 Find $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = \mathcal{L}^{-1}\left\{\underbrace{\frac{1}{s}}_{F(s)} \cdot \underbrace{\frac{1}{s^2+1}}_{G(s)}\right\} = 1 * \sin t$

$$= \int_0^t 1 \cdot \sin(t-u)\,du = \cos(t-u)\Big|_0^t = \cos 0 - \cos t = 1 - \cos t.$$

2 Find $F(s)$ of $f(t) = \int_0^t (t-u)^2 \underbrace{\cos 2u}_{g(u)}\,du = t^2 * \cos 2t$

$f(t-u) = (t-u)^2 \Rightarrow f(t) = t^2$

$$\mathcal{L}\{f(t)\} = F(s) = \frac{2}{s^3}\,\frac{s}{s^2+4}$$

3 Find $F(s)$ of $f(t) = \int_0^t \underbrace{(t-u)}_{f(t)=t} e^u\,du = t * e^t \Rightarrow g(t) = e^t.$

$$F(s) = \frac{1}{s^2}\cdot\frac{1}{s-1}$$

4 Find inverse Laplace of

a $H(s) = \frac{s}{(s+1)(s^2+4)} = \frac{1}{s+1}\cdot\frac{s}{s^2+4} \Rightarrow \mathcal{L}^{-1}\{H(s)\} = e^{-t} * \cos 2t$

$$e^{-t} * \cos 2t = \int_0^t e^{-u}\cos(2(t-u))\,du = \underbrace{\int_0^t e^{-(t-u)}\cos 2u\,du}_{\text{use integration by (parts)}}$$

(b) $H(s) = \frac{1}{s^4(s^2+1)} = \frac{3!}{3!}\frac{1}{s^4} \cdot \frac{1}{s^2+1}$

$$\mathcal{L}^{-1}\{H(s)\} = \frac{1}{6} t^3 * \sin t = \frac{1}{6}\int_0^t u^3 \sin(t-u)\,du = \frac{1}{6}\int_0^t (t-u)^3 \sin u\,du$$

(integration by parts - exercise

Ex Find the solution of $y(t)$ if

$$y(t) = 4t - 3\int_0^t y(u)\sin(t-u)\,du$$

Take Laplace transform of both sides, let $\mathcal{L}\{y(t)\} = Y(s)$

$$Y(s) = \frac{4}{s^2} - 3\mathcal{L}\{y(t) * \sin t\}$$

$$Y(s) = \frac{4}{s^2} - 3\frac{Y(s)}{s^2+1}$$

$$\left(1 + \frac{3}{s^2+1}\right)Y(s) = \frac{4}{s^2} \Rightarrow \quad Y(s) = \frac{4s^2+4}{s^2(s^2+4)}$$

$$Y(s) = \frac{\overset{0}{\overset{\shortparallel}{A}}}{s} + \frac{\overset{1}{\overset{\shortparallel}{B}}}{s^2} + \frac{\overset{0}{\overset{\shortparallel}{C}}s + \overset{3}{\overset{\shortparallel}{D}}}{s^2+4} = \frac{3}{2} \cdot \frac{1 \cdot 2}{s^2+4}$$

Laplace inverse $\quad y(t) = t + \frac{3}{2}\sin 2t.$

EXAMPLES

(1) Compute the following Laplace transforms.

① $\mathcal{L}\{1+t^2\} = \mathcal{L}\{1\} + \mathcal{L}\{t^2\} = \frac{1}{s} + \frac{3!}{s^3}$

② $\mathcal{L}\{\sin 2t - t^3\} = \mathcal{L}\{\sin 2t\} - \mathcal{L}\{t^3\} = \frac{2}{s^2+4} - \frac{3!}{s^4}$

③ $\mathcal{L}\{5 - e^{2t} + 6t^2\} = \mathcal{L}\{5\} - \mathcal{L}\{e^{2t}\} + \mathcal{L}\{6t^2\}$

$= \frac{5}{s} - \frac{1}{s-2} + 6 \cdot \frac{s^3}{3!}$

→ Rule $\mathcal{L}\{t^{n-1} e^{at}\} = F(s-a)$ $\qquad \mathcal{L}\{e^{at} f(t)\} = F(s-a)$

$F(s) = \mathcal{L}\{f(t)\}$

$\mathcal{L}\{t^{n-1}\} = \frac{(n-1)!}{s^n}$ $\qquad \mathcal{L}\{t^{n-1} e^{at}\} = F(s-a) = \frac{(n-1)!}{(s-a)^n}$

④ $\mathcal{L}\{t^3 e^{8t}\} = F(s-8) = \frac{3!}{(s-a)^4}$

$\mathcal{L}\{t^3\} = \frac{3!}{s^4}$

⑤ $\mathcal{L}\{e^{7t} \sin^2 t\} = F(s-7)$ $\qquad \sin^2 t = \frac{1}{2} - \frac{1}{2}\cos 2t$

$F(s) = \mathcal{L}\{\sin^2 t\} = \mathcal{L}\left\{\frac{1}{2} - \frac{1}{2}\cos 2t\right\} = \mathcal{L}\{1/2\} - \frac{1}{2}\mathcal{L}\{\cos 2t\}$

$= \frac{1}{2s} - \frac{1}{2}\frac{s}{s^2+4} = F(s)$

$F(s-7) = \frac{1}{2(s-7)} - \frac{1}{2}\frac{(s-7)}{(s-7)^2+4}$

⑥ $\mathcal{L}\{e^{-t}\cos 2t\} = F(s+1) = \frac{s+1}{(s+1)^2+4}$ $\qquad \mathcal{L}\{\cos 2t\} = F(s) = \frac{s}{s^2+4}$

→ Rule $\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} F(s)$ $\qquad F(s) = \mathcal{L}\{f(t)\}$

⑦ $\mathcal{L}\{t^3 \cos t\} = (-1)^3 \frac{d^3}{ds^3}[F(s)]$ $\qquad F(s) = \mathcal{L}\{\cos t\} = \frac{s}{s^2+1}$

$= -1 \frac{d^3}{ds^3}\left(\frac{s}{s^2+1}\right) = \frac{6s^4 - 36s^2 + 6}{(1+s^2)^4}$

(8) $\mathcal{L}\{t^2 \sin t\} = (-1)^2 \frac{d^2}{ds^2}[F(s)]$ $\quad [F(s)] = \mathcal{L}\{\sinh at\}$

$$= \frac{a}{s^2-a^2}.$$

$$= \frac{d^4}{ds^4}\left[\frac{a}{s^2-a^2}\right]$$

$$= \frac{24(5as^4+10a^3s^2-a^5)}{(s^2-a^2)^5}.$$

(2) Compute inverse transformations of the following functions

→ Rule $\quad \mathcal{L}^{-1}\{C_1 F_1(s) + C_2 F_2(s)\} = C_1 \mathcal{L}^{-1}\{F_1(s)\} + C_2 \mathcal{L}^{-1}\{F_2(s)\}$

$$= C_1 f_1(t) + C_2 f_2(t)$$

(1) $F(s) = \frac{3}{s+4}$ $\quad \mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{3}{s+4}\right\} = 3e^{-4t}$

(2) $F(s) = \frac{8s}{s^2+16}$ $\quad \mathcal{L}^{-1}\{F(s)\} = 8\,\mathcal{L}^{-1}\left\{\frac{s}{s^2+16}\right\} = 8\cos 4t$

(3) $F(s) = \frac{3s-12}{s^2+8}$ $\quad \mathcal{L}^{-1}\left\{\frac{3s-12}{s^2+8}\right\} = 3 \cdot \mathcal{L}^{-1}\left\{\frac{s}{s^2+8}\right\} - \frac{12}{\sqrt{8}}\mathcal{L}^{-1}\left\{\frac{\sqrt{8}}{s^2+8}\right\}$

$$= 3\cos(\sqrt{8}t) - \frac{12}{\sqrt{8}}\sin\sqrt{8}t$$

(4) $F(s) = \frac{1}{s^2-2s+9} = \frac{1}{(s-1)^2+8}$ $\quad \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^2+8}\right\} = \frac{1}{\sqrt{8}}\mathcal{L}^{-1}\left\{\frac{\sqrt{8}}{(s-1)^2+8}\right\}$

$$= \frac{1}{\sqrt{8}}e^{x}\sin\sqrt{8}x$$

(5) $F(s) = \frac{s+4}{(s+2)^2+4} = \frac{s+2}{(s+2)^2+4} + \frac{2}{(s+2)^2+4}$

$$\mathcal{L}^{-1}\{F(s)\} = \mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+4}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{(s+2)^2+4}\right\}.$$

$$= e^{-2t}\cos 2t + e^{-2t}\sin 2t.$$

$\mathcal{L}\{e^{at}\cos bt\} = \frac{s-a}{(s-a)^2+b^2}.$ $\quad \mathcal{L}\{e^{at}\sin bt\} = \frac{b}{(s-a)^2+b^2}.$

③ Use Laplace transformation to solve

$$y''(t) - y'(t) - 6y(t) = 2 \qquad y(0) = 1 \quad y'(0) = 0$$

$$\mathcal{L}\{y''(t)\} - \mathcal{L}\{y'(t)\} - 6\mathcal{L}\{y(t)\} = 2\mathcal{L}\{1\}$$

Rule $\mathcal{L}\{f^{(n)}(t)\} = s^n \mathcal{L}\{f(t)\} - s^{n-1}f(0) \cdots - sf^{(n-2)}(0) - f^{(n-1)}(0)$

$$s^2\mathcal{L}\{y(t)\} - s\underbrace{y(0)}_{1} - \underbrace{y'(0)}_{0} - [s\mathcal{L}\{y(t)\} - y(0)] - 6\mathcal{L}\{y(t)\} = \frac{2}{s}$$

$$(s^2 - s - 6)\mathcal{L}\{y(t)\} = \frac{2}{s} + s - 1 = \frac{s^2 - s + 2}{s}$$

$$\mathcal{L}\{y\} = \frac{s^2 - s + 2}{s(s^2 - s - 6)} = -\frac{1}{3s} + \frac{8}{15}\frac{1}{s-3} + \frac{4}{5}\frac{1}{s+2}$$

$$y(t) = -\frac{1}{3} + \frac{8}{15}e^{3t} + \frac{4}{5}e^{-2t}$$

④ Evaluate following Laplace transformations using convolution

① $\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} - ?$

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

$$g(t) = \mathcal{L}^{-1}\{G(s)\}$$

$$\frac{1}{s(s^2+1)} = \underbrace{\frac{1}{s}}_{F(s)} \; \underbrace{\frac{1}{s^2+1}}_{G(s)}$$

$$\mathcal{L}^{-1}\{1/s\} = 1 \qquad \mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} = \sin t$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = \int_0^x 1 \cdot \sin(x-t)\,dt = \cos(x-t)\Big|_0^x$$

or $$\mathcal{L}^{-1}\left\{\frac{1}{s(s^2+1)}\right\} = \int_0^x f(x-t)g(t)\,dt = \int_0^x \sin t\,dt = -\cos t\Big|_0^x$$

$$= 1 - \cos x$$

(2) $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\} = ?$

$\mathcal{L}^{-1}\{F(s)\} = t = f(t)$

$\frac{1}{s^2(s^2+1)} = \frac{1}{s^2} \cdot \frac{1}{s^2+1}$ $\quad F(s) \quad G(s)$

$\mathcal{L}^{-1}\{G(s)\} = \sin t = g(t)$

(I) $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\} = \int^x f(t)g(x-t)\,dt = \int^x t\sin(x-t)\,dt.$

$= \int_0^x t\,[\sin x\cos t - \cos x\sin t]\,dt = \sin x\int_0^x t\cos t\,dt - \cos x\int_0^x t\sin t$

$= \sin x\,[x\sin x + \cos x - 1] - \cos x\,[-x\cos x + \sin x]$

$\int t\cos t\,dt = t\cdot\sin t - \int \sin t\,dt = t\sin t + \cos t$

$t = u \quad \cos t\,dt = dv$

$dt = du \quad \sin t = v$

$\int t\sin t\,dt = -t\cos t + \int\cos t\,dt = -t\cos t + \sin t$

$t = u \quad \sin t\,dt = dv$

$dt = du \quad -\cos t = v$

(II) $\mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+1)}\right\} = \int_0^x g(t)f(x-t)\,dt = \int_0^x \sin t\,(x-t)\,dt.$

$= x\int_0^x \sin t\,dt - \int_0^x \sin t\cdot t\,dt = -x\cos t\Big|_0^x - [\sin t - t\cos t]\Big|_0^x$

$= -x\cos x + x - [\sin x - x\cos x] = \cancel{-x\cos x} + x - \sin x + \cancel{x\cos x} = x - \sin x$

(3) Use laplace to solve $y' - 3y = 4e^{5t}$ $\quad y(0) = 6$

$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$

$\mathcal{L}\{y'\} - 3\mathcal{L}\{y\} = 4\mathcal{L}\{e^{5t}\}$

$s\,\mathcal{L}\{y\} - 6 - 3\mathcal{L}\{y\} = \frac{4}{s-5}$

$\mathcal{L}\{y\}(s-3) = \frac{4}{s-5} + 6$

$$\mathcal{L}\{y\} = \frac{4}{(s-5)(s-3)} + \frac{6}{s-3}$$

$$\frac{4}{(s-5)(s-3)} = \frac{A}{s-5} + \frac{B}{s-3}$$

$A(s-3) + B(s-5) = 4$

$A = -B \qquad -3A - 5B = 4$

$3B - 5B = 4 \qquad -2B = 4$

$B = -2$

$A = 2.$

$$\mathcal{L}\{y\} = \frac{2}{s-5} - \frac{2}{s-3} + \frac{6}{s-3} = \frac{2}{s-5} + \frac{4}{s-3}$$

$$y(x) = \mathcal{L}^{-1}\left\{\frac{2}{s-5}\right\} + \mathcal{L}^{-1}\left\{\frac{4}{s-3}\right\} \qquad y(x) = 2e^{5x} + 4\cdot e^{3x}$$

④ $y' + y = x \qquad y(0) = 1.$

$$\mathcal{L}\{y' + y\} = \mathcal{L}\{x\} \qquad \underbrace{s\mathcal{L}\{y\} - y(0)}_{\mathcal{L}\{y'\}} + \mathcal{L}\{y\} = \frac{1}{s^2}$$

$$\mathcal{L}\{y\}(s+1) - 1 = \frac{1}{s^2} \qquad \mathcal{L}\{y\}(s+1) = \frac{1}{s^2} + 1$$

$$\mathcal{L}\{y\} = \frac{s^2+1}{(s+1)s^2} \qquad \mathcal{L}\{y\} = \frac{1}{s+1} + \frac{1}{s^2(s+1)}$$

$$= \frac{1}{s+1} + \frac{-s+1}{s^2} + \frac{1}{s+1}$$

$$\frac{s^2+1}{(s+1)s^2} = \frac{As+B}{s^2} + \frac{D}{s+1}$$

$(s+1) \qquad s^2$

$$= \frac{2}{s+1} - \frac{1}{s} + \frac{1}{s^2}$$

$As^2 + As + Bs + B + Ds^2 = 1$

$(A+D)s^2 + (A+B)s + B = 1$

$B = 1 \qquad A = -1 \quad D = 1$

$$y = \underbrace{\mathcal{L}^{-1}\left\{\frac{2}{s+1}\right\}} - \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s}\right\}}_{1} + \underbrace{\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}}_{x}$$

$$\boxed{y(x) = 2e^{-x} + x - 1.}$$

(5) $y'' + k^2 y = x \qquad y(0) = 1 \qquad y'(0) = -2$

$$\mathcal{L}\{y''\} + k^2 \mathcal{L}\{y\} = \mathcal{L}\{x\}$$

$$\underbrace{s^2 \mathcal{L}\{y\} - s\,\underbrace{y(0)}_{1} - \underbrace{y'(0)}_{-2}}_{\mathcal{L}\{y''\}} + k^2 \mathcal{L}\{y\} = \frac{1}{s^2}$$

$$(s^2 + k^2)\,\mathcal{L}\{y\} = \frac{1}{s^2} + s - 2$$

$$\mathcal{L}\{y\} = \frac{1}{s^2(s^2+k^2)} + \frac{s}{s^2+k^2} - \frac{2}{s^2+k^2}$$

$$y(x) = \mathcal{L}^{-1}\left\{\frac{1}{s^2(s^2+k^2)}\right\} + \mathcal{L}^{-1}\left\{\frac{s}{s^2+k^2}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{s^2+k^2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\cdot\frac{1}{s^2+k^2}\right\} = \frac{1}{k}\int_0^x t \sin k(x-t)\,dt = \frac{x}{k^2} - \frac{\sin kx}{k^3}$$

$$y(x) = \frac{x}{k^2} - \frac{\sin(kx)}{k^3} + \cos(kx) - \frac{2}{k}\sin(kx)$$

(5) $y''' - y'' + y' - y = 0 \qquad y(0) = 1 \qquad y'(0) = 0$

$$\mathcal{L}\{y'''\} - \mathcal{L}\{y''\} + \mathcal{L}\{y'\} - \mathcal{L}\{y\} = 0$$

$$s^3 \mathcal{L}\{y\} - s^2 y(0) - s y'(0) - y''(0) - s^2 \mathcal{L}\{y\} + s y(0) + y'(0)$$

$$+ s\,\mathcal{L}\{y\} - y(0) - \mathcal{L}\{y\} = 0$$

$$\mathcal{L}\{y\}\,(s^3 - s^2 + s - 1) = s^2 - s + 1$$

$$\mathcal{L}\{y\} = \frac{s^2 - s + 1}{s^3 - s^2 + s - 1} = \frac{s^2 - s + 1}{(s-1)(s^2+1)} = \frac{1}{2(s-1)} + \frac{s-1}{2(s^2+1)}$$

$$y(x) = \frac{1}{2}e^x + \frac{1}{2}(\cos x - \sin x)$$

$u_c(t) = \begin{cases} 0 & t<c \\ 1 & t\geq c \end{cases}$ $\qquad \mathcal{L}\{u_c(t)\} = \frac{e^{-sc}}{s}$ $\qquad \mathcal{L}\{u_c(t) f(t-c)\} = e^{-cs}F(s)$

$F(s) = \mathcal{L}\{f(t)\}$

Define $g(t) = f(t-c)$

$$\mathcal{L}\{g(t)\, u_c(t)\} = \mathcal{L}\{f(t-c)\, u_c(t)\} = e^{-cs} F(s) = e^{-cs}\mathcal{L}\{f(t)\}$$
$$= e^{-cs}\mathcal{L}\{g(t+c$$

(1) $$\mathcal{L}\{t^2 \cdot u_1(t)\} = e^{-s}\mathcal{L}\{(t+1)^2\} = e^{-s}\mathcal{L}\{t^2+2t+1\}$$
$$= e^{-s}(\mathcal{L}\{t^2\} + 2\mathcal{L}\{t\} + \mathcal{L}\{1\})$$
$$= e^{-s}\left[\frac{2}{s^3} + \frac{2}{s^2} + \frac{1}{s}\right]$$

(2) $$\mathcal{L}^{-1}\left\{\frac{e^{-2s}}{s^2}\right\} = (t-2)\cdot u_c(t) = \begin{cases} 0 & t<2 \\ t-2 & t\geq 2. \end{cases}$$

$F(s) = \frac{1}{s^2} \qquad c=2 \qquad \mathcal{L}\{\frac{1}{s^2}\} = t.$

(3) write $f(t) = \begin{cases} 2 & 0<t<1 \\ t^2/2 & 1<t<\pi/2 \\ \cos t & t>\pi/2 \end{cases}$ using Heaviside function and obtain its laplace

$u_1(t) = \begin{cases} 0 & 0<t<1 \\ 1 & t\geq 1. \end{cases}$ $\qquad u_{\pi/2}(t) = \begin{cases} 0 & 0<t<\pi/2 \\ 1 & t\geq \pi/2. \end{cases}$

$$f(t) = 2 + \begin{cases} 0 & 0<t<1 \\ \frac{t^2}{2}-2 & 1<t<\pi/2 \\ t^2/2 - 2 & t>\pi/2 \end{cases} + \begin{cases} 0 & 0<t<1 \\ 0 & 1<t<\pi/2 \\ (\cos t - t^2/2) & t>\pi/2 \end{cases}$$

$(\frac{t^2}{2}-t)\, u_1(t) = \begin{cases} 0 & 0<t<1 \\ \frac{t^2}{2}-2 & t>1 \end{cases}$ $\qquad (\cos t - t^2/2)\, u_{\pi/2}(t) = \begin{cases} 0 & 0<t< \\ \cos t - \frac{t^2}{2} & t\geq \end{cases}$

$$f(t) = \begin{cases} 2 & 0<t<1 \\ t^2/2 - \cancel{2} + \cancel{2} & 1<t<\pi/2 \\ \cancel{t^2/2} - \cancel{2} + \cos t - \cancel{t^2/2} + \cancel{2} & t>\pi/2 \end{cases}$$

$$f(t) = 2 + \left(\frac{t^2}{2}-2\right) u_1(t) + \left(\cos t - \frac{t^2}{2}\right) u_{\pi/2}(t)$$

$$\mathcal{L}\{f(t)\} = \mathcal{L}\{2\} + \mathcal{L}\{(\tfrac{t^2}{2}-2)\, u_1(t)\} + \mathcal{L}\{(\cos t-\tfrac{t^2}{2})\, u_{\pi/2}(t)\}$$

$$= \frac{1}{s} +$$

$$\mathcal{L}\{f(t-a)u_a(t)\} = e^{-as}F(s) \qquad F(s) = \mathcal{L}\{f(t)\}$$

$$\mathcal{L}^{-1}\{e^{-as}F(s)\} = f(t-a)u_a(t)$$

$$\mathcal{L}\{g(t)\, u_a(t)\} = \mathcal{L}\{\underbrace{f(t-a)}_{\text{we will generate}}\, u_a(t)\}$$

$$t^2 u_1(t) = (t-1)^2 \cdot u_1(t) + 2\cdot(t-1)\cdot u_1(t) + u_1(t)$$

$$t^2 = (t-1)^2 + 2(t-1) + 1$$

$$t^2 = (t-\pi/2)^2 + \pi(x-\pi/2) + \pi^2/4$$

$$(t^2-\pi/2)^2 + \pi t + \frac{\pi^2}{4} + \frac{\pi^2}{4} - \pi^2/4$$

$$t^2 u_{\pi/2}(t) = (t-\pi/2)^2 u_{\pi/2}(t) + \pi(t-\pi/2)u_{\pi/2}(t) + \frac{\pi^2}{4}u_{\pi/2}(t)$$

$$\cos t = -\sin(t-\pi/2) \qquad t > \pi/2$$

$$\mathcal{L}\{f(x)\} = \mathcal{L}\{2\} + \frac{1}{2}\Big[\mathcal{L}\{(t-1)u_1(t)\} + 2\{(t-1)u_1(t)\}$$

$$+ \mathcal{L}\{u_1(t)\}\Big] - 2\mathcal{L}\{u_1(t)\} - \mathcal{L}\{\sin(t-\pi/2)\, u_{\pi/2}(t)\}$$

$$-\frac{1}{2}\{\mathcal{L}((t-\pi/2)^2 u(t-\pi/2))\} + \pi\,\mathcal{L}\{(t-\pi/2)u(\pi/2)\}$$

$$+ \frac{\pi^2}{4}\mathcal{L}\{u(t-\pi/2)\}$$

$$\mathcal{L}\{f(t)\} = \frac{2}{s} - 2e^{s} + \Big[\frac{1}{s^3} + \frac{1}{s^2} + \frac{1}{2s}\Big]e^{-s}$$

$$-\Big(\frac{1}{s^3} + \frac{\pi}{2s^2} + \frac{\pi^2}{8s}\Big)e^{-\pi s/2} - \frac{1}{s^2+1}e^{-s\pi/2}$$

(4) Solve $y''+16y=g(t)$ $\quad y(0)=0 \quad y'(0)=1$

$$g(t)=\begin{cases}\cos 4t & 0\le t<\pi \\ 0 & t\geq \pi\end{cases}$$

$$g(t) = \cos 4t - \cos 4t\, U_{\pi}(t)$$
$$= \cos 4t - \cos 4(t-\pi)\, U_{\pi}(t)$$

$$\mathcal{L}^{-1}\{e^{-as}F(s)\}=f(t-a)U_a(t)$$

$$\mathcal{L}\{y''\} + 16\,\mathcal{L}\{y\} = \mathcal{L}\{g(t)\}$$

$$s^2\mathcal{L}\{y\} - \cancel{s\,y(0)}^{\,0} - y'(0) + 16\,\mathcal{L}\{y\} = \frac{s}{s^2+16} - \frac{s}{s^2+16}e^{-\pi s}$$

$$(s^2+16)\,\mathcal{L}\{y\} = 1+\frac{s}{s^2+16} - \frac{s}{s^2+16}e^{-\pi s}$$

$$\mathcal{L}\{y\} = \frac{1}{s^2+16} + \frac{s}{(s^2+16)^2} - \frac{s}{(s^2+16)^2}e^{-\pi s}$$

$$y(t) = \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{4}{s^2+16}\right\} + \frac{1}{8}\mathcal{L}^{-1}\left\{\frac{8s}{(s^2+16)^2}\right\} - \frac{1}{8}\mathcal{L}^{-1}\left\{\frac{\overset{F(s)}{8s}\,e^{-\pi s}}{(s^2+16)^2}\right\}$$

$$= \frac{1}{4}\sin 4t + \frac{1}{8}t\sin 4t - \frac{1}{8}(t-\pi)\sin 4(t-\pi)\,U_{\pi}(t)$$

$$y(t) = \begin{cases}\frac{1}{4}\sin 4t + \frac{1}{8}t\sin 4t & 0\le t<\pi \\ \frac{2+\pi}{8}\sin 4t & t\geq\pi\end{cases}$$

$$U_{\pi}(t) = \begin{cases}0 & t<\pi \\ 1 & t\geq\pi\end{cases}$$

(5) $y''+y = \delta(t-2\pi)$ $\quad y(0)=0 \quad y'(0)=1$

$$\mathcal{L}\{y''\}+\mathcal{L}\{y\} = \mathcal{L}\{\delta(t-2\pi)\}$$

$$s^2\mathcal{L}\{y\} - s\underbrace{y(0)}_{0} - \underbrace{y'(0)}_{1} + \mathcal{L}\{y\} = e^{-2\pi s}$$

$$(s^2+1)\,\mathcal{L}\{y\} = e^{-2\pi s}+1 \qquad \mathcal{L}\{y\} = \frac{1}{s^2+1} + \frac{e^{-2\pi s}}{s^2+1}$$

$$y(t) = \sin t + \sin(t-2\pi)\,U_{2\pi}$$

Table of Laplace Transforms

	$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$		$f(t)=\mathfrak{L}^{-1}\{F(s)\}$	$F(s)=\mathfrak{L}\{f(t)\}$
1.	1	$\frac{1}{s}$	2.	$\mathbf{e}^{at}$	$\frac{1}{s-a}$
3.	$t^n,\quad n=1,2,3,\ldots$	$\frac{n!}{s^{n+1}}$	4.	$t^p, p>-1$	$\frac{\Gamma(p+1)}{s^{p+1}}$
5.	$\sqrt{t}$	$\frac{\sqrt{\pi}}{2s^{\frac{3}{2}}}$	6.	$t^{n-\frac{1}{2}},\quad n=1,2,3,\ldots$	$\frac{1\cdot 3\cdot 5\cdots(2n-1)\sqrt{\pi}}{2^n s^{n+\frac{1}{2}}}$
7.	$\sin(at)$	$\frac{a}{s^2+a^2}$	8.	$\cos(at)$	$\frac{s}{s^2+a^2}$
9.	$t\sin(at)$	$\frac{2as}{\left(s^2+a^2\right)^2}$	10.	$t\cos(at)$	$\frac{s^2-a^2}{\left(s^2+a^2\right)^2}$
11.	$\sin(at)-at\cos(at)$	$\frac{2a^3}{\left(s^2+a^2\right)^2}$	12.	$\sin(at)+at\cos(at)$	$\frac{2as^2}{\left(s^2+a^2\right)^2}$
13.	$\cos(at)-at\sin(at)$	$\frac{s\left(s^2-a^2\right)}{\left(s^2+a^2\right)^2}$	14.	$\cos(at)+at\sin(at)$	$\frac{s\left(s^2+3a^2\right)}{\left(s^2+a^2\right)^2}$
15.	$\sin(at+b)$	$\frac{s\sin(b)+a\cos(b)}{s^2+a^2}$	16.	$\cos(at+b)$	$\frac{s\cos(b)-a\sin(b)}{s^2+a^2}$
17.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	18.	$\cosh(at)$	$\frac{s}{s^2-a^2}$
19.	$\mathbf{e}^{at}\sin(bt)$	$\frac{b}{(s-a)^2+b^2}$	20.	$\mathbf{e}^{at}\cos(bt)$	$\frac{s-a}{(s-a)^2+b^2}$
21.	$\mathbf{e}^{at}\sinh(bt)$	$\frac{b}{(s-a)^2-b^2}$	22.	$\mathbf{e}^{at}\cosh(bt)$	$\frac{s-a}{(s-a)^2-b^2}$
23.	$t^n\mathbf{e}^{at},\quad n=1,2,3,\ldots$	$\frac{n!}{(s-a)^{n+1}}$	24.	$f(ct)$	$\frac{1}{c}F\left(\frac{s}{c}\right)$
25.	$u_c(t)=u(t-c)$ Heaviside Function	$\frac{\mathbf{e}^{-cs}}{s}$	26.	$\delta(t-c)$ Dirac Delta Function	$\mathbf{e}^{-cs}$
27.	$u_c(t)f(t-c)$	$\mathbf{e}^{-cs}F(s)$	28.	$u_c(t)g(t)$	$\mathbf{e}^{-cs}\mathfrak{L}\{g(t+c)\}$
29.	$\mathbf{e}^{ct}f(t)$	$F(s-c)$	30.	$t^n f(t),\quad n=1,2,3,\ldots$	$(-1)^n F^{(n)}(s)$
31.	$\frac{1}{t}f(t)$	$\int_s^{\infty}F(u)\,du$	32.	$\int_0^t f(v)\,dv$	$\frac{F(s)}{s}$
33.	$\int_0^t f(t-\tau)g(\tau)\,d\tau$	$F(s)G(s)$	34.	$f(t+T)=f(t)$	$\frac{\int_0^T \mathbf{e}^{-st}f(t)\,dt}{1-\mathbf{e}^{-sT}}$
35.	$f'(t)$	$sF(s)-f(0)$	36.	$f''(t)$	$s^2F(s)-sf(0)-f'(0)$
37.	$f^{(n)}(t)$	$s^nF(s)-s^{n-1}f(0)-s^{n-2}f'(0)\cdots-sf^{(n-2)}(0)-f^{(n-1)}(0)$			

Table Notes

1. This list is not a complete listing of Laplace transforms and only contains some of the more commonly used Laplace transforms and formulas.

2. Recall the definition of hyperbolic functions.

$$\cosh(t) = \frac{\mathbf{e}^t + \mathbf{e}^{-t}}{2} \qquad \sinh(t) = \frac{\mathbf{e}^t - \mathbf{e}^{-t}}{2}$$

3. Be careful when using "normal" trig function vs. hyperbolic functions. The only difference in the formulas is the "+ a^2" for the "normal" trig functions becomes a "- a^2" for the hyperbolic functions!

4. Formula #4 uses the Gamma function which is defined as

$$\Gamma(t) = \int_0^\infty \mathbf{e}^{-x} x^{t-1}\, dx$$

If n is a positive integer then,

$$\Gamma(n+1) = n!$$

The Gamma function is an extension of the normal factorial function. Here are a couple of quick facts for the Gamma function

$$\Gamma(p+1) = p\Gamma(p)$$

$$p(p+1)(p+2)\cdots(p+n-1) = \frac{\Gamma(p+n)}{\Gamma(p)}$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$$